

***Generation Interconnection
System Impact Study Report***

For

***PJM Generation Interconnection Request
Queue Position AB2-091***

Hudson 230kV

April 2017

Preface

The intent of the System Impact Study is to determine a plan, with approximate cost and construction time estimates, to connect the subject generation interconnection project to the PJM network at a location specified by the Interconnection Customer. As a requirement for interconnection, the Interconnection Customer may be responsible for the cost of constructing: Network Upgrades, which are facility additions, or upgrades to existing facilities, that are needed to maintain the reliability of the PJM system. All facilities required for interconnection of a generation interconnection project must be designed to meet the technical specifications (on PJM web site) for the appropriate transmission owner.

In some instances an Interconnection Customer may not be responsible for 100% of the identified network upgrade cost because other transmission network uses, e.g. another generation interconnection or merchant transmission upgrade, may also contribute to the need for the same network reinforcement. The possibility of sharing the reinforcement costs with other projects may be identified in the Feasibility Study, but the actual allocation will be deferred until the System Impact Study is performed.

The System Impact Study estimates do not include the feasibility, cost, or time required to obtain property rights and permits for construction of the required facilities. The project developer is responsible for the right of way, real estate, and construction permit issues. For properties currently owned by Transmission Owners, the costs may be included in the study.

General

PSEG Fossil, LLC, the Interconnection Customer (IC), has proposed a natural gas generating facility located in Middlesex, New Jersey. The installed facilities will have a total capability of 105 MW with 100 MW of this output being recognized by PJM as capacity. The AB2-091 project is reclaiming CIRs from the Hudson unit. The proposed in-service date for this project is June 1, 2019. **This study does not imply a PSE&G commitment to this in-service date.**

Point of Interconnection

AB2-091 will interconnect with the PSE&G transmission system via a new breaker position at Hudson 230 kV substation

Cost Summary

The AB2-091 project will be responsible for the following costs:

Description	Total Cost
Attachment Facilities	\$ 0
Direct Connection Network Upgrades	\$ 3,567,900
Non Direct Connection Network Upgrades	\$ 0
Allocation for New System Upgrades	\$ 0
Contribution for Previously Identified Upgrades	\$ 0
Total Costs	\$ 3,567,900

Attachment Facilities

No Attachment Facilities are required to support this interconnection request.

Direct Connection Cost Estimate

The total preliminary cost estimate for the Direct Connection work is given in the table below. These costs do not include CIAC Tax Gross-up.

Description	Total Cost
Hudson 230kV Substation reconfigure the bay position by relocating F-2311 to other side of the tower and associated buswork, UG cables, switches, relaying, and metering. PJM Network Upgrade #N5267	\$ 3,567,900
Total Direct Connection Cost Estimate	\$ 3,567,900

Non-Direct Connection Cost Estimate

No Non-Direct Connection Facilities are required to support this interconnection request.

Transmission Owner Scope of Work

- Relocate F-2311 to other side of the tower that is near Hudson 230kV BS 10
- Install (per phase) new 230kV disconnect and associated ground on the existing A-frame above Hudson 230kV BS 10
- Install foundation for new Riser Pole and new Riser Pole structure with mounted overhead to underground terminations
- Install ~250 ft. of strain bus per phase between A-frame above BS 10 and Riser Pole.
- Install ~400 ft. of underground XLPE cable per phase from Riser Pole terminations to manhole outside the station. (Point of interconnection is assumed to be the manhole outside the station).
- Install associated Relays, Line Protection, Metering, and ESOC racks.

Schedule

Based on the extent of the Direct Connection and system upgrades required to support the AB2-091 generation project, it is expected to take a minimum of 36 months from the date of a fully executed Interconnection Construction Service Agreement to complete the installation.

Revenue Metering and SCADA Requirements

PJM Requirements

The Interconnection Customer will be required to install equipment necessary to provide Revenue Metering (KWH, KVARH) and real time data (KW, KVAR) for IC's generating Resource. See PJM Manuals M-01 and M-14D, and PJM Tariff Sections 24.1 and 24.2.

Public Service Electric and Gas Requirements

The Interconnection Customer will be required to comply with all PSE&G Revenue Metering Requirements for Generation Interconnection Customers. The Revenue Metering Requirements may be found within the "Information and Requirements for Electric Service" document located at the following links:

http://www.pseg.com/business/builders/new_service/before/
<http://www.pjm.com/planning/design-engineering/to-tech-standards.aspx>

Network Impacts

The Queue Project AB2-091 was evaluated as a 105.0 MW (Capacity 100.0 MW) injection at the Hudson 230kV substation in the PSEG area. Project AB2-091 was evaluated for compliance with applicable reliability planning criteria (PJM, NERC, NERC Regional Reliability Councils, and Transmission Owners). Project AB2-091 was studied with a commercial probability of 100%. Potential network impacts were as follows:

Summer Peak Analysis - 2020

Generator Deliverability

(Single or N-1 contingencies for the Capacity portion only of the interconnection)

None.

Multiple Facility Contingency

(Double Circuit Tower Line contingencies were studied for the full energy output. The contingencies of Line with Failed Breaker and Bus Fault will be performed for the Impact Study.)

None.

Short Circuit

(Summary of impacted circuit breakers)

None.

Contribution to Previously Identified Overloads

(This project contributes to the following contingency overloads, i.e. "Network Impacts", identified for earlier generation or transmission interconnection projects in the PJM Queue)

None.

Steady-State Voltage Requirements

(Summary of the VAR requirements based upon the results of the steady-state voltage studies)

None.

Stability and Reactive Power Requirement for Low Voltage Ride Through

(Summary of the VAR requirements based upon the results of the dynamic studies)

See Attachment 3 for analysis

New System Reinforcements

(Upgrades required to mitigate reliability criteria violations, i.e. Network Impacts, initially caused by the addition of this project generation)

None.

Contribution to Previously Identified System Reinforcements

(Overloads initially caused by prior Queue positions with additional contribution to overloading by this project. This project may have a % allocation cost responsibility which will be calculated and reported for the Impact Study)

None.

Potential Congestion due to Local Energy Deliverability

PJM also studied the delivery of the energy portion of this interconnection request. Any problems identified below are likely to result in operational restrictions to the project under study. The developer can proceed with network upgrades to eliminate the operational restriction at their discretion by submitting a Merchant Transmission Interconnection request.

Note: Only the most severely overloaded conditions are listed below. There is no guarantee of full delivery of energy for this project by fixing only the conditions listed in this section. With a Transmission Interconnection Request, a subsequent analysis will be performed which shall study all overload conditions associated with the overloaded element(s) identified

None.

Light Load Analysis:

Contribution to Previously Identified Overloads

(PECO-BGE) The PEACHBTM – CNASTONE 500kV line (from bus 200013 to bus 200004 ckt 1) loads from 112% to 115% of its normal rating 2338 MVA for the Base Case non-contingency condition. AB2-091 project contributes approximately 29.10 MW to the overload.

To relieve the Peach Bottom - Conastone 500kV line overload:

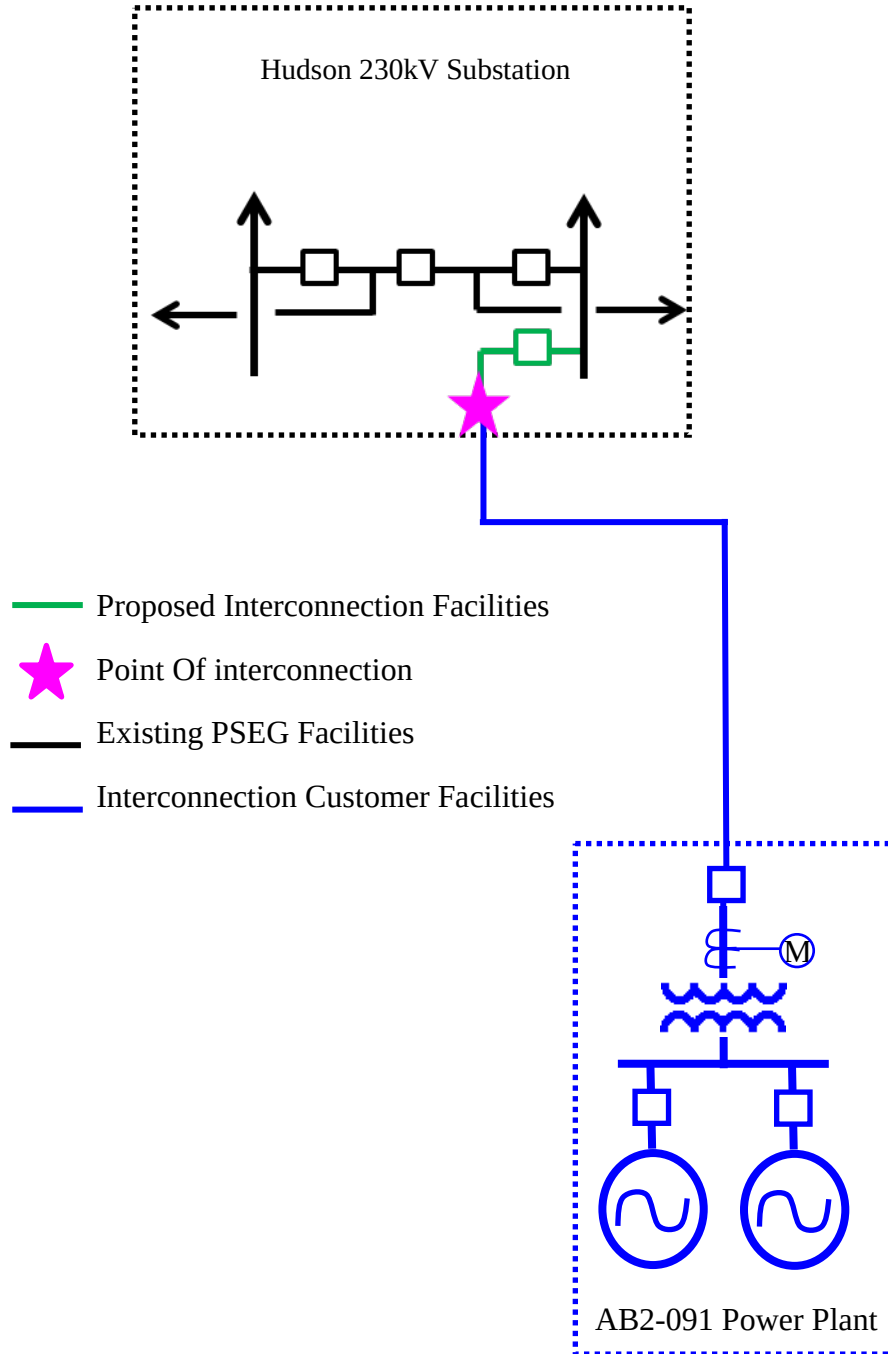
Existing baseline B2766.1; upgrade substation equipment at Peach Bottom 500 kV (on the Peach Bottom – Conastone 500 kV circuit) to increase facility rating to 2826 MVA normal and 3525 MVA emergency. AB2-091 will require B2766.1 to be advanced from 2021 to 2020. AB2-091 is responsible for the cost required to advance B2766.2 from 2021 to 2020.

Cost: PECO advancement cost = \$0.

Existing baseline B2766.2; Upgrade substation equipment at Conastone 500 kV (on the Peach Bottom – Conastone 500 kV circuit) to increase facility rating to 2826 MVA normal and 3525 MVA emergency. AB2-091 will require B2766.2 to be advanced from 2021 to 2020. AB2-091 is responsible for the cost required to advance B2766.2 from 2021 to 2020.

Cost: BGE advancement cost = \$0.

Attachment 1. Single Line Diagram



Attachment 2. Dynamic Simulation Analysis

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Executive Summary

Generator Interconnection Request AB2-091 is for a 105 MW Maximum Facility Output (MFO) natural gas combustion turbine facility. AB2-091 is a 2 X 76.9 MVA BRUSH BDAX 7-290ERJT generator with a Point of Interconnection (POI) connected to Hudson 230kV switchyard in the PSEG transmission system, Hudson, NJ.

This report describes a dynamic simulation analysis of AB2-091 as part of the overall system impact study.

The load flow scenario for the analysis was based on the RTEP 2020 Light Load case, modified to include applicable queue projects. AB2-091 has been dispatched online at maximum power output, with approximately unity power factor at the POI.

The AB2-091 queue project was tested for compliance with NERC, PJM and other applicable criteria. The range of contingencies evaluated was limited to that necessary to assess compliance and each was limited to a 20-second simulation time period.

Simulated NERC Standard TPL-001 faults include:

1. Three-phase (3ph) fault with normal clearing (Category P1)
2. Operating of a line section w/o a fault, Single-line-to-ground (slg) on Bus Section and Breaker. (Category P2)
3. Single-line-to-ground (slg) with delayed clearing as a result of breaker failure (Category P4)
4. Single-line-to-ground (slg) with delayed clearing as a result of protection failure (Category P5)
5. Single-line-to-ground (slg) with normal clearing for common structure (Category P7)

Note: For generator interconnection studies, Category P3 and P6 faults will be studied on an as needed basis. In this study, P2 contingencies are not applicable.

Other applicable criteria tested include:

1. Transmission Owner (TO) specific criteria
2. Other criteria

The system was tested for a system intact condition and the fault types listed above. Specific fault descriptions and breaker clearing times used for this study are provided in the result table.

No relevant High Speed Reclosing (HSR) contingencies were identified.

For all simulations, the queue project under study along with the rest of the PJM system were required to maintain synchronism and with all states returning to an acceptable new condition following the disturbance.

For the remaining fault contingencies tested on the 2020 Light Load case:

- a) Post-contingency oscillations were positively damped with a damping margin of at least 4% for local modes and 3% for inter-area modes.
- b) The AB2-091 generator was able to ride through all faults (except for faults where protective action trips a generator(s)).
- c) Following fault clearing, all bus voltages recover to a minimum of 0.7 per unit after 2.5 seconds (except where protective action isolates that bus).
- d) No transmission element trips, other than those either directly connected or designed to trip as a consequence of that fault.

No mitigations were found to be required.

1. Introduction

Generator Interconnection Request AB2-091 is for is for a 105 MW Maximum Facility Output (MFO) natural gas combustion turbine facility. AB2-091 is a 2 X 76.9 MVA BRUSH BDAX 7-290ERJT generator with a Point of Interconnection (POI) connected to Hudson 230kV switchyard in the PSEG transmission system, Hudson, NJ..

This analysis is effectively a screening study to determine whether the addition of AB2-091 will meet the dynamic requirements of the NERC, PJM and Transmission Owner reliability standards.

In this report the AB2-091 project and how it is proposed to be connected to the grid are first described, followed by a description of how the project is modeled in this study. The fault cases are then described and analyzed, and lastly a discussion of the results is provided.

2. Description of Project

AB2-091 consists of 2 x 55 MW Combustine Turbine generators. The two generators will be connected to the POI via a 230/13.8/13.8 kV three windings generator step up (GSU) transformer with a rating of 75/100/125 MVA. The AB2-091 Point of Interconnection (POI) is the Hudson 230kV substation circuit as shown in Figure 1.

Table 1 lists the parameters given in the impact study data and the corresponding parameters of the AB2-091 loadflow models.

The dynamic model for the AB2-091 plant is based on the model data supplied by the Developer.

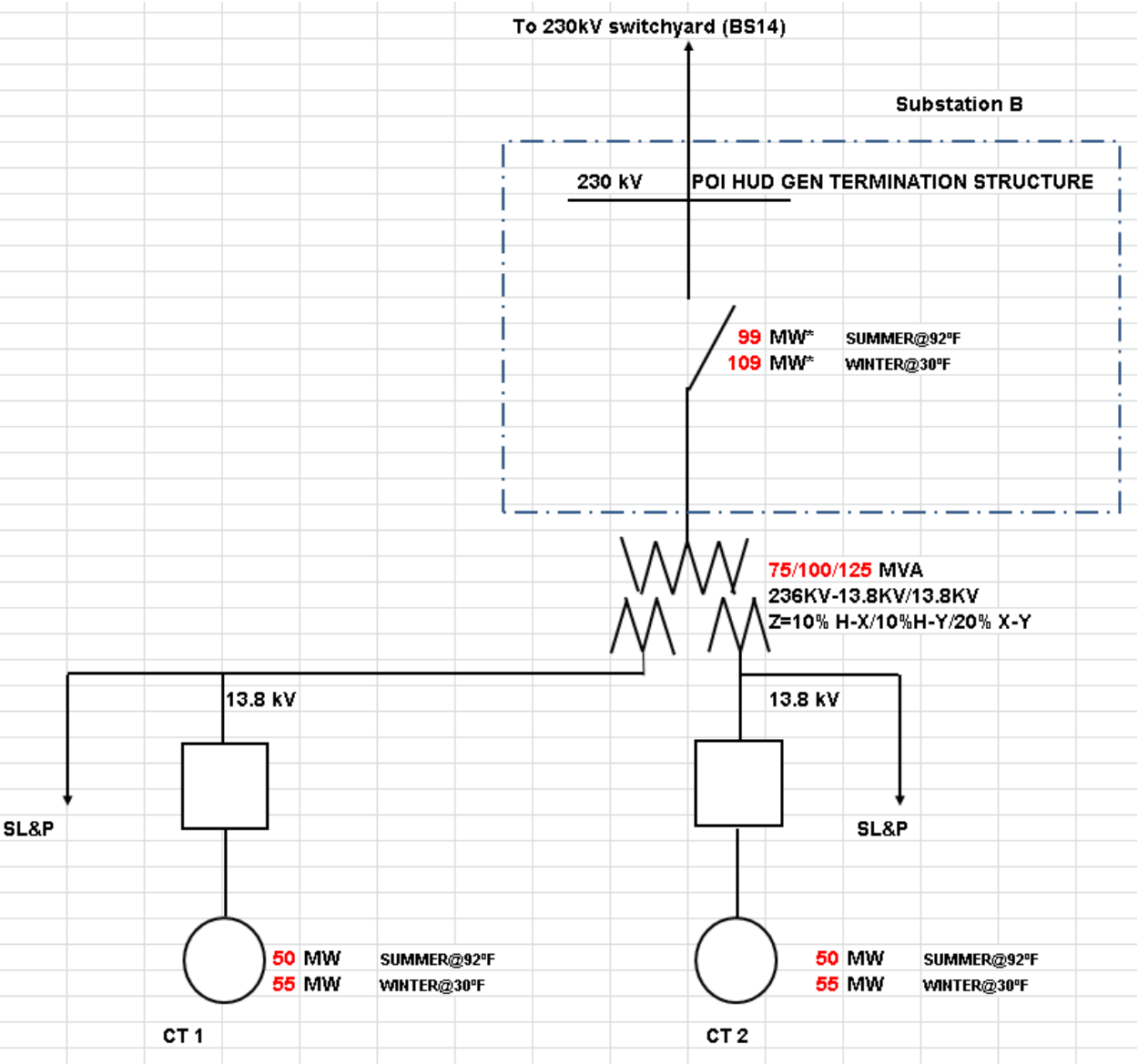


Figure 1: AB2-091 Plant Model

Table 1: AB2-091 Plant Model

	Impact Study Data	Model
Generators	2 x 65.4 MW BDAX 7-290ERJT Generator MVA base = 76.9 MVA V_t = 13.8 kV Unsaturated sub-transient reactance = 0.195 pu @ rated MVA	2 X 55MW generator Pgen 55 MW Pmax 55 MW Pmin 0 MW Qmax 26.64 MVA_r Qmin -18.08 MVA_r Mbase 76.94 MVA Zsorce j0.195 pu @ Mbase
GSU transformers	1 x 236/13.8/13.5 kV two winding transformers Rating = 75/100/125 MVA Transformer base = 75 MVA Impedance: H-X = 0.1 pu @ MVA base H-Y = 0.1 pu @ MVA base X-Y = 0.2 pu @ MVA base Number of taps = 5 Tap step size = 2.5	1 x 230/13.8/13.5 kV two winding transformers Rating = 75/100/125 MVA Transformer base = 75 MVA Impedance: H-X = 0.1 pu @ MVA base H-Y = 0.1 pu @ MVA base X-Y = 0.2 pu @ MVA base Number of taps = 5 Tap step size = 2.5 Norminal tap = 2.5
Auxiliary load	1.0 MW + 0.6 MVAR	0.5 MW + 0.3 MVAR at low voltage side of each GSU

Station load	1.0 MW + 0.6 MVAR	0.5 MW + 0.3 MVAR at low voltage side of each GSU
Transmission line	Not Provided	ZBR

3. Loadflow and Dynamics Case Setup

The dynamics simulation analysis was carried out using PSS/E Version 33.7.

The load flow scenario and fault cases for this study are based on PJM's Regional Transmission Planning Process¹.

The selected load flow scenario is the RTEP 2020 Light Load case with the following modifications:

- a) Addition of all applicable queue projects prior to AB2-091.
- b) Addition of AB2-091 queue project.
- c) Removal of withdrawn and subsequent queue projects in the vicinity of AB2-091.
- d) Dispatch of units in the PJM system to maintain slack generators within limits.

The AB2-091 initial conditions are listed in Table 2, indicating maximum power output, with AB2-091 regulating to unity power factor at the generator bus.

Table 2: AB2-091 machine initial conditions

Bus	Name	Unit	PGEN (MW)	QGEN (MVAR)	ETERM (p.u.)	POI Voltage (p.u.)
924422	AB2-091 1	1	55	-3.3	1.0000	1.0323
924423	AB2-091 2	1	55	-3.3	1.0000	1.0323

Generation within the vicinity of AB2-091 has been dispatched online at maximum output (P_{MAX}). The dispatch of generation in the vicinity of AB2-091 is given in Attachment 3.

¹ Manual 14B: PJM Region Transmission Planning Process, Rev 33, May 5 2016, Attachment G : PJM Stability, Short Circuit, and Special RTEP Practices and Procedures.

4. Fault Cases

Tables 3 listed the contingencies and results that were studied, with representative worst case total clearing times provided by PJM. Each contingency was studied over a 20 second simulation time interval.

Simulated NERC Standard TPL-001 faults include:

1. Three-phase (3ph) fault with normal clearing (Category P1)
2. Operating of a line section w/o a fault, Single-line-to-ground (slg) on Bus Section and Breaker. (Category P2)
3. Single-line-to-ground (slg) with delayed clearing as a result of breaker failure (Category P4)
4. Single-line-to-ground (slg) with delayed clearing as a result of protection failure (Category P5)
5. Single-line-to-ground (slg) with normal clearing for common structure (Category P7)

Note: For generator interconnection studies, Category P3 and P6 faults will be studied on an as needed basis. In this study, P2 contingencies are not applicable.

Other applicable criteria tested include:

1. Transmission Owner (TO) specific criteria
2. Other criteria

The system was tested for a system intact condition and the fault types listed above. No relevant High Speed Reclosing (HSR) contingencies were studied.

5. Evaluation Criteria

This study is focused on AB2-091, along with the rest of the PJM system, maintaining synchronism and having all states return to an acceptable new condition following the disturbance. The recovery criteria applicable to this study are as per PJM's Regional Transmission Planning Process and Transmission Owner criteria:

- a) The system with AB2-091 included is transiently stable and post-contingency oscillations should be positively damped with a damping margin of at least 4% for local modes and 3% for inter-area modes.
- b) The AB2-091 is able to ride through faults (except for faults where protective action trips AB2-091).
- c) Following fault clearing, all bus voltages recover to a minimum of 0.7 per unit after 2.5 seconds (except where protective action isolates that bus).
- d) No transmission element trips, other than those either directly connected or designed to trip as a consequence of that fault.

6. Summary of Results

Plots from the dynamic simulations are provided in Attachment 4, with results summarized in Table 3.

Due to the frequency protection was disabled due to the PSSE deficiency in calculating frequencies.

For the fault contingencies tested in this study:

- a) Post-contingency oscillations were positively damped with a damping margin of at least 4% for local modes and 3% for inter-area modes.
- b) The AB2-091 generator was able to withstand all contingencies.
- c) Following fault clearing, all bus voltages recover to a minimum of 0.7 per unit after 2.5 seconds (except where protective action isolates that bus).
- d) No transmission element trips, other than those either directly connected or designed to trip as a consequence of that fault.

7. Mitigations

No Mitigations were found to be required.

Table 3: Fault list**P0: Steady State**

Fault ID	Duration
P0.00	Steady State 20 sec run

P1: Three Phase Faults with normal clearing

Fault ID	Fault description	Clearing Time Normal (Cycles)	Results
P1.00	3ph @ AB2-091 POI – Hudson 230kV line, normal clear	5	Stable
P1.01	3ph @ Hudson – Kearny 230kV line circuit F-2311, normal clear	5	Stable
P1.02	3ph @ Hudson – Penhorn Y 230kV line, normal clear	5	Stable
P1.03	3ph @ Hudson – Penhorn X – Homestead – North Bergen 230kV line, normal clear	5	Stable
P1.04	3ph @ Hudson – South Waterfront 230kV line circuit H-2313, normal clear	5	Stable
P1.05	3ph @ Hudson – South Waterfront 230kV line circuit P-2268, normal clear	5	Stable
P1.06	3ph @ Hudson – Kearny 230kV line circuit G2312, normal clear	5	Stable
P1.07	3ph @ Hudson – Essex 230kV line, normal clear	5	Stable
P1.08	3ph @ Hudson – Belleville 230kV line, normal clear	5	Stable
P1.09	3ph @ South Waterfront (BS2) – Newport 230kV line circuit L2264, normal clear	5	Stable
P1.10	3ph @ South Waterfront (BS7) – Newport 230kV line circuit Q2269, normal clear	5	Stable
P1.11	3ph @ Newport – Hoboken 230kV line circuit I-2340 (#1), normal clear	5	Stable
P1.12	3ph @ Newport – Hoboken 230kV line circuit T-2220 (#2), normal clear	5	Stable
P1.13	3ph @ Hoboken – 49 St – North Bergen – Bergen 230kV line, normal clear	5	Stable

P4: SLG Stuck Breaker (SB) Faults at Backup Clearing

Fault ID	Fault description	Clearing Time Normal/Delayed (Cycles)	Results
P4.01	SLG @ Hudson – Kearny 230kV line, SB @ Hudson, delayed clear loss of Hudson 230/13kV Tx	5/19	Stable
P4.02	SLG @ Hudson – Penhorn Y 230kV line, SB @ Hudson, delayed clear loss of Hudson reactor	5/19	Stable
P4.03	SLG @ Hudson – Penhorn Y 230kV line, SB @ Hudson, delayed clear loss of Hudson reactor	5/19	Stable
P4.04	SLG @ Hudson – Penhorn – Homestead – North Bergen 230kV line, SB @ Hudson, delayed clear loss of AB2-091	5/19	Stable
P4.05	SLG @ Hudson – South Waterfront 230kV line circuit H-2313, SB @ Hudson, delayed clear loss of Hudson reactor	5/19	Stable
P4.06	SLG @ South Waterfront – Hudson 230kV line circuit H-2313, SB @ South Waterfront, delayed clear loss of South Waterfront (BS7) – Newport 230kV line circuit Q-2269	5/19	Stable
P4.07	SLG @ South Waterfront (BS2) – Newport 230kV line circuit L2264, SB @ South Waterfront, delayed clear loss of South Waterfront – Hudson 230kV line circuit P-2268	5/19	Stable
P4.08	SLG @ Hudson – Belleville 230kV line, SB @ Hudson, delayed clear loss of Hudson – Kearny 230kV line circuit G2312	5/19	Stable
P4.09	SLG @ Newport – South Waterfront (BS7) 230kV line circuit Q-2269, SB @ Newport, delayed clear loss of Newport – Hoboken 230kV line circuit I-2340 (#1)	5/19	Stable

P5: SLG Fault with Delayed (Zone 2) Clearing

Fault ID	Fault description	Clearing Time Normal/Delayed (Cycles)	Results
P5.01	SLG @ Hudson – Kearny 230kV line circuit F-2311, relay failure @ Kearny, delayed clear	5/34	Stable
P5.02	SLG @ Kearny – Hudson 230kV line circuit F-2311, relay failure @ Hudson, delayed clear	5/34	Stable
P5.03	SLG @ Hudson – Penhorn 230kV line, relay failure @ PenHorn, delayed clear	5/34	Stable
P5.04	SLG @ Penhorn – Hudson 230kV, relay failure @ Hudson, delayed clear	5/34	Stable

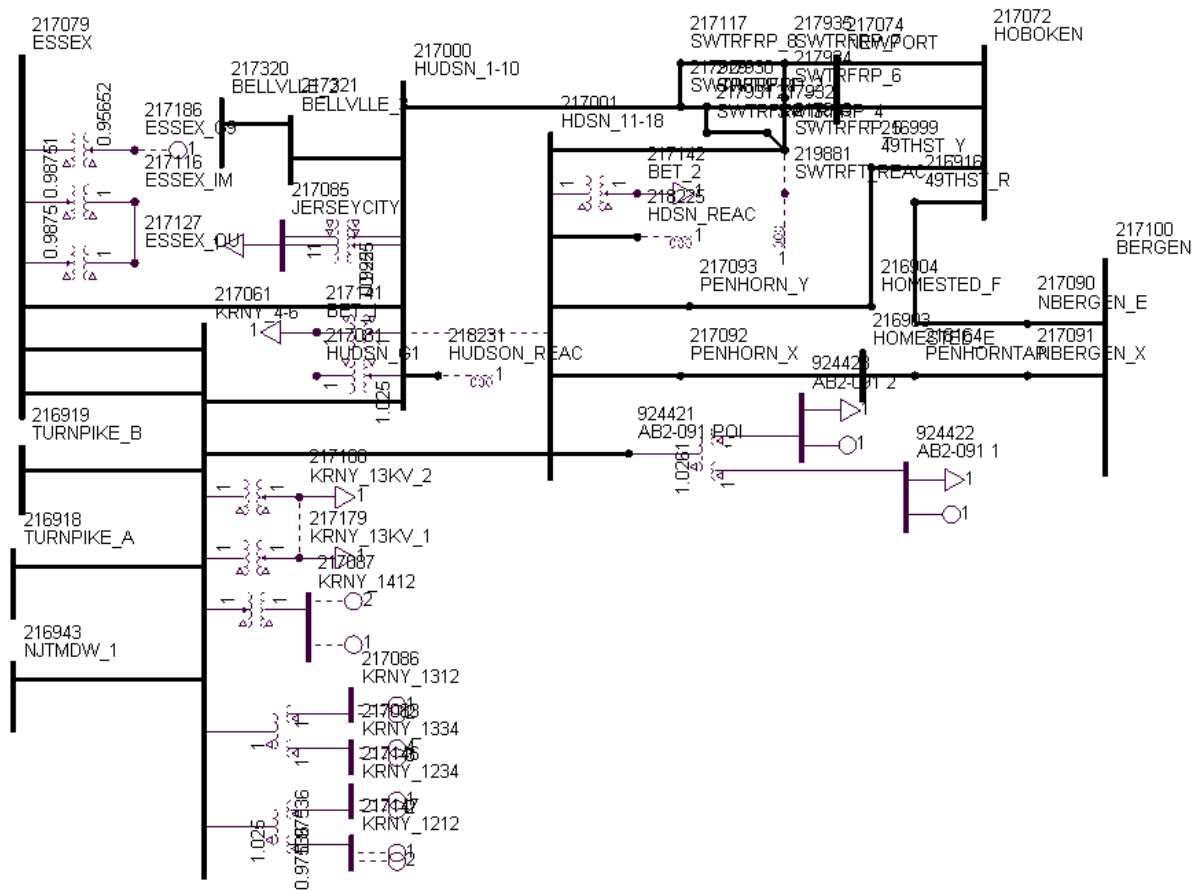
Fault ID	Fault description	Clearing Time Normal/Delayed (Cycles)	Results
P5.05	SLG @ Hudson – Penhorn_X– Homestead – North Bergen 230kV line, relay failure @ North Bergen, delayed clear	5/34	Stable
P5.06	SLG @ North Bergen – Homestead – Penhorn_X – Hudson 230kV line, relay failure @ Hudson, delayed clear	5/34	Stable
P5.07	SLG @ Hudson – South Waterfront 230kV line circuit H-2313, relay failure @ South Waterfront, delayed clear	5/34	Stable
P5.08	SLG @ South Waterfront – Hudson 230kV line circuit H-2313, relay failure @ Hudson, delayed clear	5/34	Stable
P5.09	SLG @ Hudson – South Waterfront 230kV line circuit P-2268, relay failure @ South Waterfront, delayed clear	5/34	Stable
P5.10	SLG @ South Waterfront – Hudson 230kV line circuit P-2268, relay failure @ Hudson, delayed clear	5/34	Stable
P5.11	SLG @ Hudson – Kearny 230kV line circuit G2312, relay failure @ Kearny, delayed clear	5/34	Stable
P5.12	SLG @ Kearny – Hudson 230kV line circuit G2312, relay failure @ Hudson, delayed clear	5/34	Stable
P5.13	3ph @ Essex – Hudson 230kV line, relay failure @ Hudson, delayed clear	5/34	Stable
P5.14	3ph @ Belleville – Hudson 230kV line, relay failure @ Hudson, delayed clear	5/34	Stable
P5.15	3ph @ Newport – South Waterfront (BS2) 230kV line circuit L2264, relay failure @ South Waterfront (BS2), delayed clear	5/34	Stable
P5.16	3ph @ Newport – South Waterfront (BS7) 230kV line circuit Q2269, relay failure @ South Waterfront (BS7), delayed clear	5/34	Stable

P7: Common Structure

Fault ID	Fault description	Clearing Time (Cycles)	Results
P7.01	CONTINGENCY 'HMSD-PNHT+R2270' SLG @ Hudson – Penhorn X – Homestead – North Bergen 230kV line, tower failure normal clear loss of Bergen – North Bergen – Hoboken 230kV line	5	Stable

Fault ID	Fault description	Clearing Time (Cycles)	Results
P7.02	CONTINGENCY 'X2250+Y2225' SLG @ Hudson – Penhorn X – Homestead – North Bergen 230kV line, tower failure normal clear loss of Hudson – Penhorn Y 230kV line	5	Stable
P7.03	CONTINGENCY 'A2227+KRY-EX' SLG @ Hudson – Essex 230kV line, tower failure normal clear loss of Kearny – Essex 230kV line #2	5	Stable
P7.04	CONTINGENCY 'HMSD-HOBK+HMSD-PNHT' SLG @ Hudson – Penhorn X – Homestead – North Bergen 230kV line, tower failure normal clear loss of Penhorn Y – 49 St – Hoboken 230kV	5	Stable

Appendix 1. PSS/E Model One Line Diagram



Appendix 2. AB2-091 PSS/E Dynamic Model

REPORT FOR PLANT MODELS

BUS 924422 [AB2-091 1 13.800] MODELS

Model GENROU Bus 924422 [AB2-091 1 13.800] Machine "1 " :

C O N S		S T A T E S									
99012-99025		41864-41869									
M B A S E		Z S O R C E		X T R A N		G E N T A P					
76.9		0.00000+J 0.19500		0.00000+J 0.00000		1.00000					
T'D0	T''D0	T'Q0	T''Q0	H	DAMP	XD	XQ	X'D	X'Q	X''D	XL
9.50	0.050	2.90	0.050	0.90	0.00	2.5400	2.3200	0.2640	0.3800	0.1950	0.1400
S(1.0)				S(1.2)							
0.1100				0.4900							

Model PSS2B Bus 924422 [AB2-091 1 13.800] Machine "1 " :

I C O N S		C O N S		S T A T E S		V A R S											
11426-11431		132669-132691		68296-68312		24060-24063											
IC1		REMBUS1		IC2		REMBUS2		M		N							
1		0		3		0		5		1							
TW1		TW2		T6		TW3		TW4		T7		KS2		KS3			
3.000		3.000		0.000		3.000		3.000		2.000		1.400		1.000			
T8		T9		KS1		T1		T2		T3		T4		T10		T11	
0.500		0.100		1.500		1.000		1.000		1.000		0.010		0.070		0.010	
VS1MAX		VS1MIN		VS2MAX		VS2MIN		VSTMAX		VSTMIN							
999.900		-999.900		999.900		-999.900		0.100		-0.100							

Model AC8B Bus 924422 [AB2-091 1 13.800] Machine "1 " :

C O N S		S T A T E S		V A R S									
244368-244388		98338-98342		29119-29121									
TR		KPR		KIR		KDR		TDR		VPIDMAX			
VPIDMIN	0.0000	84.0000		5.0000		10.0000		0.1000		999.0000		-999.0000	
KA		TA		VRMAX		VRMIN		KC		KD			
1.0000		0.0000		46.8300		0.0000		0.6700		2.0200			
KE		TE		VFEMAX		VEMIN							
1.0000		1.2000		27.8000		0.0000							
E1		S(E1)		E2		S(E2)							
7.4000		0.3200		9.9000		3.7400							

Model GGOV1 Bus 924422 [AB2-091 1 13.800] Machine "1 " :

C O N S		S T A T E S		V A R S		RES-ICONS			
334097-334129		127201-127210		56880-56900		16955-16956			
R	TPELEC	MAXERR	MINERR	KPGOV	KIGOV	KDGOV	TDGOV	VMAX	VMIN
0.050	1.000	0.023	-0.023	3.000	1.200	0.000	1.000	1.200	0.170
TACT	KTURB	WFNL	TB	TC	TENG	TFLOAD	KPLOAD	KILOAD	LDREF
0.300	1.300	0.220	0.100	0.000	0.000	0.300	1.000	3.300	1.000

ICON(M)= 1 (Feedback signal for governor droop)
ICON(M+1)= 0 (Switch for fuel source characteristic)

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Model GGOV1 Bus 924423 [AB2-091 2 13.800] Machine "1 " :
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	C O N S		S T A T E S		V A R S		RES-ICONS		
	334130-334162		127211-127220		56901-56921		16957-16958		
R	TPELEC	MAXERR	MINERR	KPGOV	KIGOV	KDGOV	TDGOV	VMAX	VMIN
0.050	1.000	0.023	-0.023	3.000	1.200	0.000	1.000	1.200	0.170
TACT	KTURB	WFNL	TB	TC	TENG	TFLOAD	KPLOAD	KILOAD	LDREF
0.300	1.300	0.220	0.100	0.000	0.000	0.300	1.000	3.300	1.000
DM	ROPEN	RCLOSE	KIMW	ASET	KA	TA	TRATE	DB	
0.000	99.000	-99.000	0.000	99.000	10.000	0.100	57.000	0.000	
TSA	TSB	RUP	RDOWN						
1.000	1.000	99.000	-99.000						

ICON(M)= 1 (Feedback signal for governor droop)

ICON(M+1)= 0 (Switch for fuel source characteristic)

Appendix 3. AB2-091 PSS/E Case Dispatch

Bus Number	Bus Name	Id	In Service	PGen (MW)	PMax (MW)	PMin (MW)	QGen (Mvar)	QMax (Mvar)	QMin (Mvar)
216901	ATHENIA_3 138.00	1	1	6.6	6.6	0	-2	2	-2
217060	ESSEX_CTG1 13.800	1	1	43	43	22	-28	20	-28
217065	ECRR_1 138.00	1	1	30.15	30.15	0	-5	10	-5
217073	ESSEX_CTG2 13.800	2	1	43	43	22	-28	20	-28
217080	ESSEX_STG3 13.800	3	1	34	34	22	-15	15	-15
217083	KRNY_1334 13.800	3	1	49.5	49.5	30	-15	22	-15
217083	KRNY_1334 13.800	4	1	49.5	49.5	30	-15	22	-15
217086	KRNY_1312 13.800	1	1	49.5	49.5	30	-15.07	21.93	-15.07
217086	KRNY_1312 13.800	2	1	49.5	49.5	30	-15.07	21.93	-15.07
217087	KRNY_1412 13.800	1	1	49.5	49.5	30	-15	22	-15
217087	KRNY_1412 13.800	2	1	49.5	49.5	30	-15	22	-15
217146	KRNY_1234 13.800	1	1	46	46	28	-15	22.15	-15
217146	KRNY_1234 13.800	2	1	46	46	28	-15	22.15	-15
217147	KRNY_1212 13.800	1	1	46	46	28	-15	22.15	-15
217147	KRNY_1212 13.800	2	1	46	46	28	-15	22.15	-15
217186	ESSEX_G9 13.800	1	1	81	81	70.47	0	0	0
217191	NEC_CT1 18.000	1	1	225.8	225.8	110.5	-59.806	158	-74.22
217192	NEC_CT2 18.000	1	1	225.8	225.8	110.5	-59.806	158	-74.22
217193	NEC_ST 18.000	1	1	240.5	240.5	51.5	63.5767	178	-79.05
217323	ECRR_2 138.00	1	1	30.15	30.15	5	-5	10	-5
292116	V1-030 CC 230.00	1	1	2.812	2.812	0	-0.928	1.35	-0.928
292191	V1-030 EC 230.00	1	1	4.588	4.588	0	-1.514	2.202	-1.514
293439	V3-066 C 230.00	1	1	0.16	0.16	0	0	0	0
293443	V3-065 C 138.00	1	1	0.21	0.21	0	0	0	0
293444	V3-065 E 138.00	1	1	0.339	0.339	0	0	0	0
910762	X3-052 E 26.000	1	1	3	3	0	-0.99	1.44	-0.99
919681	AA2-052 26.000	1	1	3	3	0	-0.99	1.44	-0.99
924422	AB2-091 1 13.800	1	1	55	55	30.5	-3.3494	26.64	-18.08
924423	AB2-091 2 13.800	1	1	55	55	30.5	-3.3494	26.64	-18.08

Appendix 4. Plots from Dynamic Simulations (See separated .PDF file)